



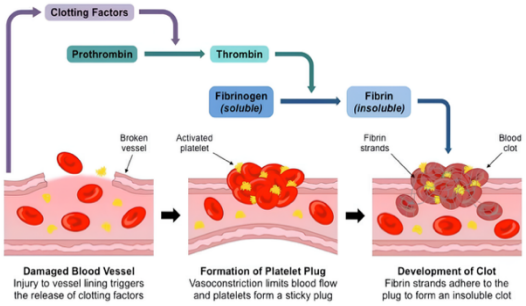

# CONTROL OF BIOLOGICAL SYSTEMS

Final project

## CLOSED-LOOP SUGGESTIONS FOR ANTICOAGULANT THERAPY


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**INTRODUCTION**




Blood coagulation is the physiological process that prevents blood loss following damage to a blood vessel. Through the coagulation cascade, a sequence of enzymatic reactions involving several clotting factors, blood changes from a liquid to a gel state, leading to the formation of fibrin threads that aggregate platelets and create a blood clot.

This process is essential for maintaining hemostasis and promoting tissue repair. The pathological formation of blood clots within blood vessels may obstruct blood flow and lead to thrombotic disorders. The most severe complications associated with thrombosis include stroke, heart attack, and serious breathing problems.

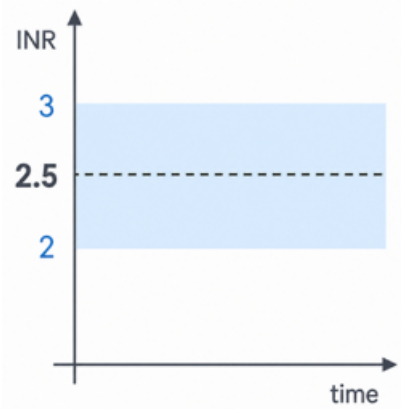
Anticoagulants are drugs that inhibit specific pathways involved in the coagulation cascade and are used to prevent or treat thrombotic disorders. This project focuses on Warfarin, an anticoagulant whose mechanism of action is based on the inhibition of vitamin K-dependent coagulation factors. By reducing the activity of these factors, Warfarin slows down the coagulation process and increases the time required for blood clot formation.

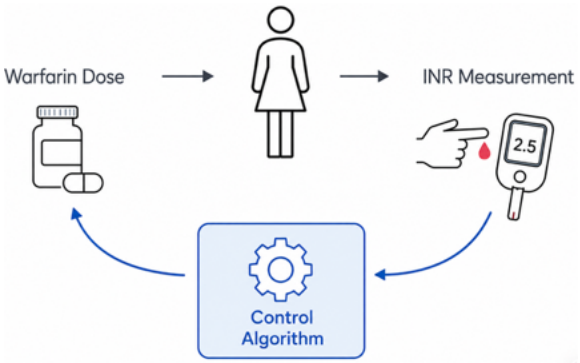
Effective Warfarin dosing is crucial because the administered dose must remain within a therapeutic range. An insufficient dose may fail to prevent clot formation, whereas an excessive dose may increase the risk of bleeding and hemorrhage. Warfarin dosing is particularly challenging because drug sensitivity can vary considerably due to the intrinsic complexity of the underlying pharmacological mechanisms and the presence of several disturbing factors.

The International Normalized Ratio INR is a standardized clinical measure of blood clotting time used to evaluate the effectiveness of anticoagulant therapy. It is obtained from the time required for a blood sample to clot, normalized with respect to an average clotting time and corrected to reduce the dependence on the specific measurement procedure. In healthy subjects, INR values are typically between 0.8 and 1.3, whereas patients undergoing Warfarin therapy usually require an INR between 2.0 and 3.0.

This project focuses on the development of closed loop control strategies for automatic Warfarin dose suggestions, with the aim of regulating the patient's INR around the therapeutic target value of 2.5. Two control approaches are designed, tuned, and tested: a Proportional Integral Derivative PID controller and a Model Predictive Control MPC controller. The controllers are required to ensure:

- $2 \leq INR(t) \leq 3$ after 5 days from the reference change ($t \geq 168\ h$)
- $INR(t) \to 2.5$ for $t \to \infty$



**TASK 1: PID control**

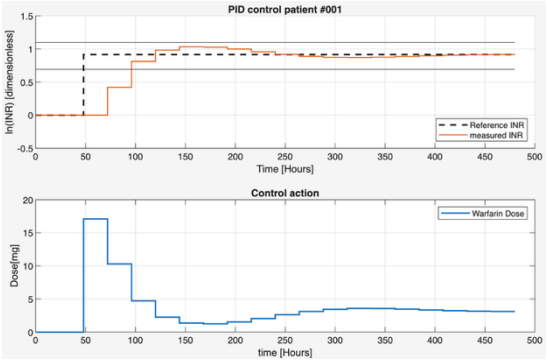
**Task 1.1: PID control of patient#001**

A discrete time PID controller was implemented and manually tuned to regulate the INR level of patient#001 under Warfarin therapy. The control objective was to maintain the INR within the therapeutic range $2 \leq INR(t) \leq 3$ after 5 days from the reference change, $t \geq 168h$, and to asymptotically track the target value INR = 2.5.

The controller operates on the tracking error expressed in the logarithmic INR domain and combines proportional, integral, and derivative actions. At each sampling instant, it computes the suggested Warfarin dose, which is saturated to keep the control action within the admissible range.
The controller parameters were manually tuned on patient#001 to satisfy the required specifications, obtaining

$$Kp = 15 \quad Ki = 0.1 \quad Kd = 30$$

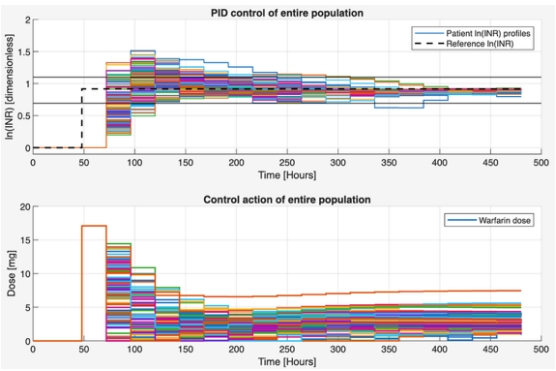
As shown in figure after the reference change at $t = 48$, the selected tuning drives the INR into the therapeutic range within the required time. A moderate overshoot is observed during the transient, while the response subsequently converges toward the target value. The Warfarin dose is initially high to rapidly increase the INR and then progressively decreases toward a maintenance value as the tracking error is reduced.

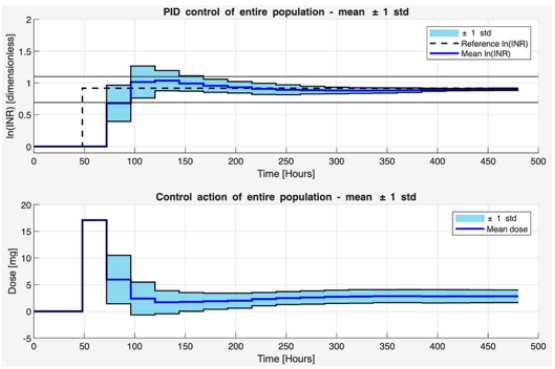



**Task 1.2: PID control of the population**

The robustness of the selected PID tuning was evaluated on the entire population of 100 patients. For each subject, the closed loop simulation was repeated using the same controller parameters selected for patient#001. Population performance was evaluated through a spaghetti plot of all individual $\ln(INR)$ trajectories and through the population mean response with its $\pm 1$ standard deviation envelope.

The results show considerable inter patient variability during the initial transient, reflecting the different responses of the patients to the same control strategy. The dispersion progressively decreases over time and the trajectories converge toward the therapeutic region. The population mean approaches the desired reference and remains within the therapeutic bounds, while the decreasing standard deviation indicates that the differences among patient responses become less pronounced as steady state is approached. The results show that the PID tuning selected on patient#001 provides satisfactory closed loop performance also when applied to the heterogeneous population.



**TASK 2: Model implementation**

**Task 2: Comparison with the nonlinear model**

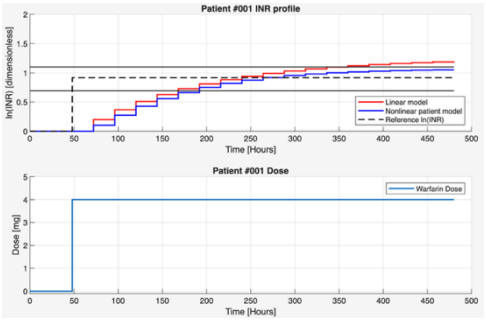
A first order discrete time linear model was implemented to approximate the response of a patient to Warfarin administration:

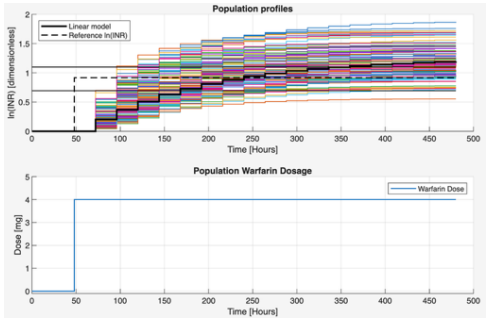
$$x(k+1) = ax(k) + bu(k), \qquad y(k) = x(k)$$

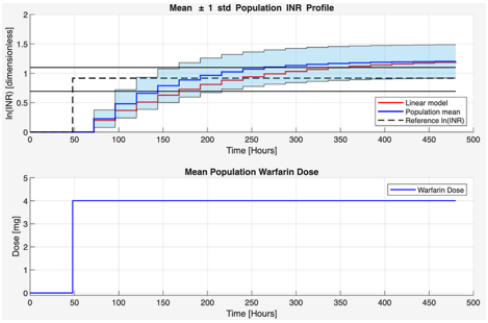
with $a = 0.84$ and $b = 0.05$ where $u(k)$ represents the Warfarin dose and the output $y(k)$ corresponds to $\ln(INR)$. Since the state coincides with the output, $x(k) = y(k)$, the state is directly accessible and no state estimator is required.

For patien#001, both the linear model and the nonlinear PK-PD simulator were subjected to the same step input: 0 mg during the first two days and 4 mg of Warfarin from day 3. The linear model reproduces the increasing trend of the nonlinear response, although a deviation between the two trajectories becomes visible over time.

The analysis was then extended to the entire population of 100 virtual patients. The plot highlights the inter patient variability of the nonlinear model, while the mean $\pm 1$ standard deviation shows that the linear trajectory follows the general trend of the population response. The selected linear model provides a representative approximation of the average patient behaviour, but it cannot reproduce the individual dynamics of each subject.



**TASK 3**

**Task 3.1: MPC for a Linear Patient Model**

A Model Predictive Controller MPC was implemented and tuned using the discrete time linear patient model introduced in Task 2. Since the state of the model coincides with its output,

$$x(k) = y(k) = \ln(INR(k)),$$

the state is directly available and no state estimator is required.

At each sampling instant, the MPC predicts the future evolution of the patient over a finite prediction horizon and computes an optimal sequence of Warfarin doses by minimizing a quadratic cost function. The optimization balances two objectives: reducing the tracking error between the predicted $\ln(INR)$ and the reference value $\ln(2.5)$, and limiting the magnitude of the control action. The physiological constraints on the administered Warfarin dose were included as

$$0 \leq u(k) \leq 50\, mg$$

The optimization problem was formulated in condensed form and solved using quadratic programming. According to the receding horizon strategy, only the first element of the optimal control sequence is applied at each sampling instant, after which the optimization is repeated using the updated patient state.

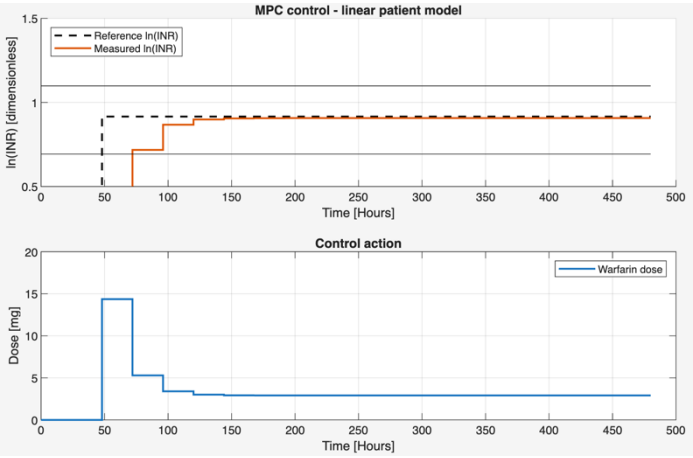
The MPC parameters were tuned on the linear model to obtain a sufficiently fast response while avoiding unnecessarily aggressive control actions. The selected values were:

- Prediction horizon:=100
- Output tracking weight: Q=10
- Control action weight: R=0.01

The relatively long prediction horizon allows the controller to account for the slow response of INR to Warfarin administration. The selected Q gives priority to reference tracking, whereas R penalizes excessive control actions.

As shown in the figure, immediately after the reference change the controller administers an initial dose of approximately 14–15 mg. The dose then rapidly decreases and progressively approaches a maintenance value of approximately 3 mg. Consequently, the $\ln(INR)$ increases toward the desired reference and enters the therapeutic region within the required 168 hours.

A small steady state tracking error remains between the output and the reference. This behaviour is expected because the MPC formulation does not include integral action. Offset free reference tracking cannot be guaranteed. This limitation motivates the introduction of integral action in Task 4.



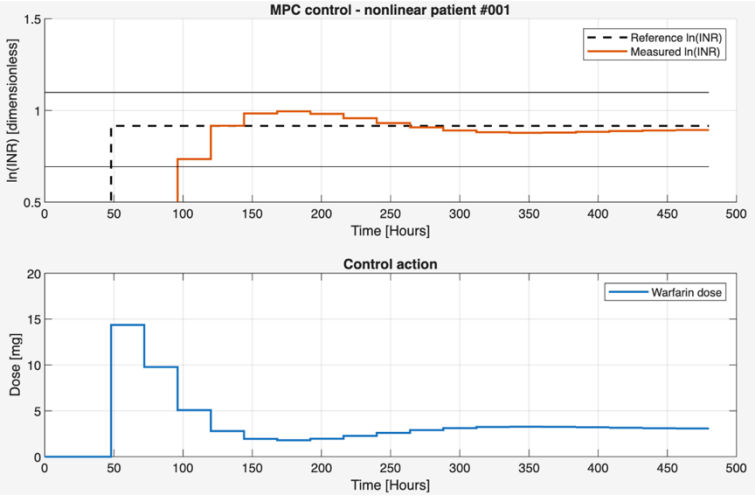
**Task 3.2: MPC for patient #001**

The MPC designed using the linear model was then applied to the nonlinear PK-PD model of patient#001, while maintaining the same tuning parameters prediction Horizon = 100, Q = 10, and R = 0.01.

Compared with the linear model, the nonlinear patient exhibits a different transient response because the prediction model used by the MPC does not exactly reproduce the actual patient dynamics. The $\ln(INR)$ initially exceeds the reference value and subsequently decreases toward it.

Despite this model mismatch, the controller successfully drives the INR into the therapeutic range within the required 168 hours. The Warfarin dose is initially high, then strongly reduced as the INR approaches the target region, and finally increases progressively toward the maintenance dose required by the nonlinear patient.

These results show that the MPC designed on the average linear model is able to regulate patient#001 despite the mismatch between the prediction model and the nonlinear plant. Residual steady state tracking error remains, consistently with the absence of integral action.



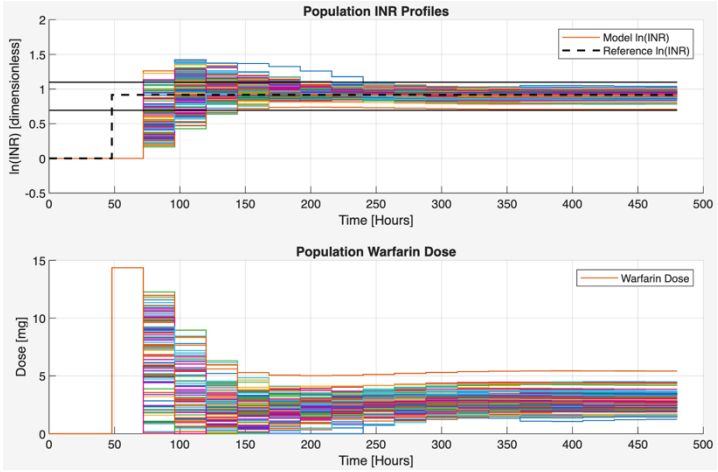

**Task 3.3: MPC for the population**

The same MPC configuration was tested on the entire population of 100 nonlinear virtual patients in order to evaluate its robustness with respect to inter patient variability. The results were represented using a spaghetti plot of all $\ln(INR)$ trajectories and Warfarin doses, together with the population mean and the corresponding $\pm 1$ standard deviation envelope.

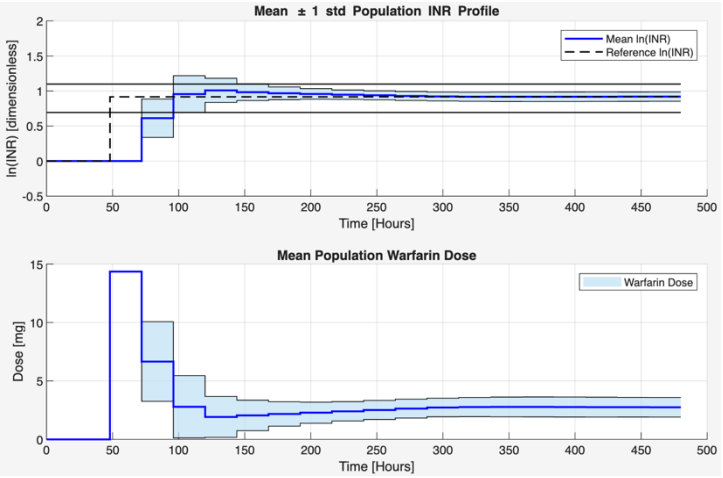
The figure shows variability during the transient phase, reflecting the different nonlinear dynamics of the individual patients. Different subjects require different control actions and exhibit different INR responses. Some trajectories show a larger overshoot, whereas others approach the reference more gradually.

The trajectories progressively move toward the therapeutic region. The population mean confirms the overall effectiveness of the controller, while the standard deviation illustrates the dispersion caused by inter patient variability. The largest differences are observed during the transient phase, when the response to the initial Warfarin administration differs most across patients.

The control actions show a similar behaviour: an high initial dose is followed by a reduction and then by convergence toward patient dependent maintenance values. At steady state, some dispersion around the reference remains. This is mainly due to the use of a single average linear prediction model to control patients characterized by different nonlinear dynamics. Since the MPC does not include integral action, offset free tracking cannot be guaranteed across the population.


Population INR Profiles
Model ln(INR)
Reference ln(INR)
ln(INR) [dimensionless]
Time [Hours]


Population Warfarin Dose
Warfarin Dose
Dose [mg]
Time [Hours]


Mean ± 1 std Population INR Profile
Mean ln(INR)
Reference ln(INR)
ln(INR) [dimensionless]
Time [Hours]


Mean Population Warfarin Dose
Warfarin Dose
Dose [mg]
Time [Hours]

**TASK 4**

**Task 4.1: MPC with Integral Action, no constraints**

To overcome the steady state tracking error observed in Task 3, the MPC controller was reformulated by introducing integral action through the full increment velocity form. The control variable is expressed in terms of the input increment

$$\Delta u(k) = u(k) - u(k-1)$$

and the linear prediction model is augmented by defining the state

$$\tilde{x}(k) = \begin{bmatrix} \Delta x(k) \\ y(k) \end{bmatrix}$$

Starting from the linear model introduced in Task 2, the augmented system is therefore described by

$$\tilde{x}(k+1) = \begin{bmatrix} A & 0 \\ CA & I \end{bmatrix} \tilde{x}(k) + \begin{bmatrix} B \\ CB \end{bmatrix} \Delta u(k)$$

with

$$y(k) = [0 \quad I]\tilde{x}(k)$$

At each sampling instant, the MPC determines the optimal sequence of future control increments by minimizing the quadratic cost function. Since the input constraints are neglected in this task, the optimization problem is unconstrained and can be solved directly. According to the receding horizon strategy, only the first optimal increment is applied and the actual Warfarin dose is reconstructed from the previous control action as

$$u(k) = u(k-1) + \Delta u(k)$$

The controller was tuned on patient#001 to obtain a sufficiently fast response while limiting excessive variations of the Warfarin dose. The selected parameters were:

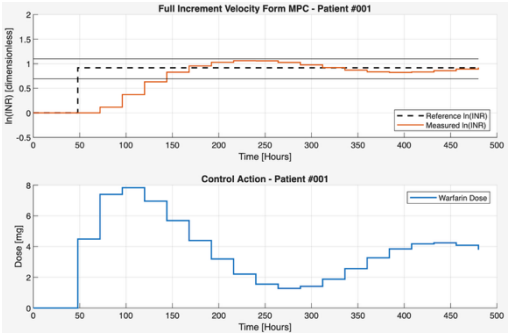
- prediction Horizon:=100
- Q: =10
- R:=0.2

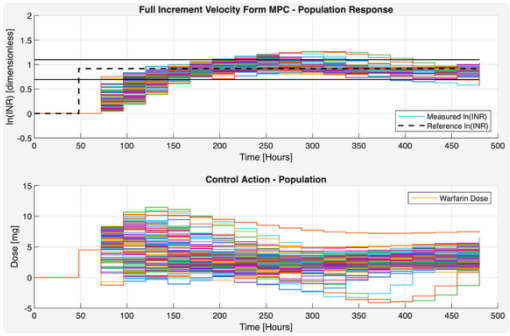
The value of R was increased with respect to Task 3 because the incremental formulation resulted in a more aggressive control action. The selected value provides a suitable compromise between tracking performance and smoothness of the dose profile.

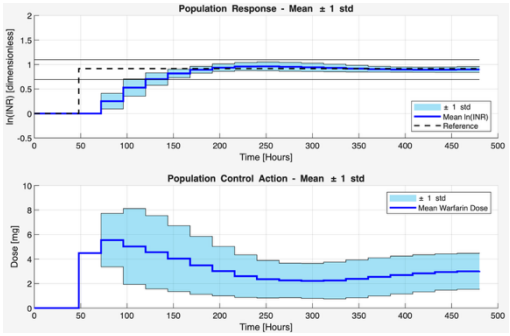
As shown in the figure, after the reference change the controller progressively increases the INR and drives patient#001 into the therapeutic range within the required 168 hours. A moderate overshoot with respect to the reference is observed during the transient, after which the controller adjusts the Warfarin dose and the response progressively approaches the desired value $\ln(2.5)$. Compared with the MPC developed in Task 3, the introduction of integral action substantially reduces the steady state tracking error.

The same controller was then tested on the entire population of nonlinear patients. The figure shows variability during the transient phase, due to the different individual responses to Warfarin. The trajectories progressively approach the therapeutic region and the population mean converges toward the reference. The standard deviation envelope also becomes narrower with time, indicating a reduction of the dispersion around the mean response.

An important limitation is visible in the population control actions. Since no constraints are imposed in Task 4.1,the MPC computes negative Warfarin doses for some patients. These values are mathematically admissible for the unconstrained optimization but cannot be physically administered, motivating the introduction of the positivity constraint in Task 4.2.




**Task 4.2: MPC with Integral Action, with constraints**

The Full Increment Velocity Form MPC developed in Task 4.1 was then extended by introducing the constraint required for the administered Warfarin dose:

$$u(k) \geq 0$$

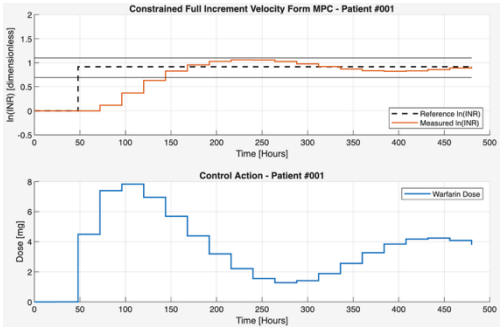
Since the optimization variable is the control increment $\Delta u(k)$, the constraint must be expressed in terms of the corresponding absolute dose. The future input sequence is obtained by accumulating the control increments from the previously applied dose. The positivity condition is included directly in the MPC optimization problem, which becomes a constrained quadratic programming problem and is solved at each sampling instant using quadprog.

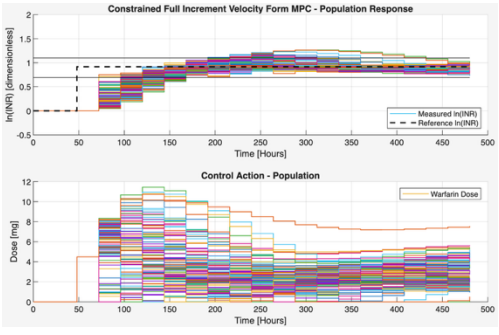
The same tuning selected in Task 4.1 was maintained: prediction Horizion = 100, Q = 10, R = 0.2

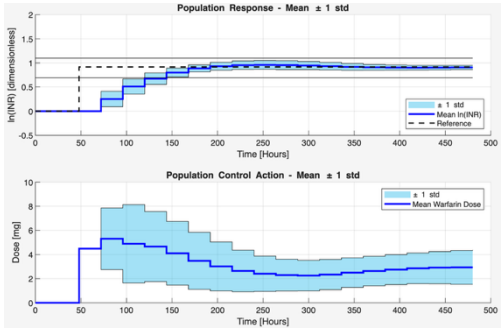
For patient#001, the constrained controller produces a response practically identical to the unconstrained case. The INR enters the therapeutic range within the required 168 hours and subsequently approaches the reference value. This occurs because the optimal Warfarin dose for patient#001 is already positive in Task 4.1, and therefore the newly introduced constraint does not become active for this patient.

The effect of the constraint becomes evident when the controller is applied to the complete population. In Task 4.1, some individual control trajectories become negative, whereas in the constrained formulation all Warfarin doses remain greater than or equal to zero. The overall INR behaviour remains very similar: the population trajectories progressively approach the therapeutic region and the mean response converges toward $\ln(2.5)$, while the standard deviation decreases toward steady state.

The constrained Full Increment Velocity Form MPC therefore preserves the tracking performance obtained with integral action while preventing non physical negative control actions, providing a more realistic formulation for Warfarin administration.



**CONCLUDING COMMENTS**

The project showed how different control strategies can be applied to the Warfarin dosing problem, progressively improving the regulation of INR. The PID controller provided satisfactory results despite the large inter patient variability, while the MPC approach allowed the future patient response and the control effort to be explicitly considered in the optimization. The introduction of integral action through the Full Increment Velocity Form improved reference tracking, and the constrained formulation ensured physically feasible Warfarin doses by preventing negative control actions.

One of the main limitations of the project is the use of a single linear model for controller design, while the actual patient population is described by nonlinear models with significantly different dynamics. This model mismatch explains part of the variability observed in the population responses and highlights the difficulty of defining a single controller that provides the same performance for all patients.

Some difficulties were encountered during the implementation of Task 4, mainly related to the Full Increment Velocity Form and to the correct implementation of the input constraint in terms of the absolute Warfarin dose. Particular attention was required to reconstruct the control action from its increments and to correctly handle the previous state and input between consecutive sampling instants. Once these aspects were addressed, the constrained MPC provided the expected behaviour.

The results show that the proposed controllers are able to regulate INR around the therapeutic target for a heterogeneous population, while also highlighting the importance of considering patient variability, model mismatch and physically meaningful constraints when designing control strategies for drug administration.